

Building ML Controlled Instruments

Goals:

- Develop a framework to unify automation of instruments
- Avoid insular solutions with individual server-client interfaces
- Flexible solution to be ready for advances in AI and ML tools

Motivation and Advantages of pyTango

- Proxy server:
 - Coherent API for clients
 - Asynchronous server
 - Enables edge computing = Smart Proxy Server
 - Dynamic device-database
 - Model—context server (MCP) compatible to interface with large language model
 - Compatible to Bluesky-Project (orchestration – system)
 - Runs more than 10k devices at synchrotrons

But: Many Manufacturers!

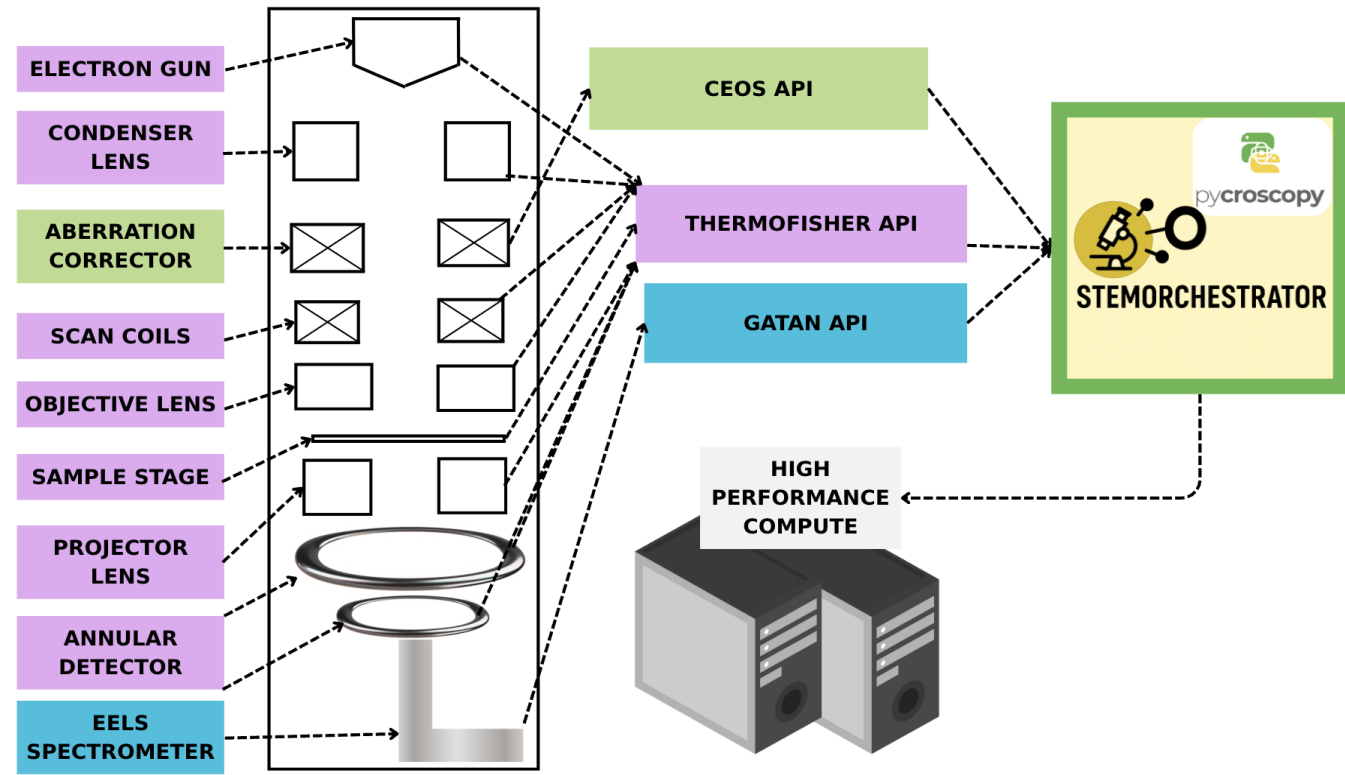
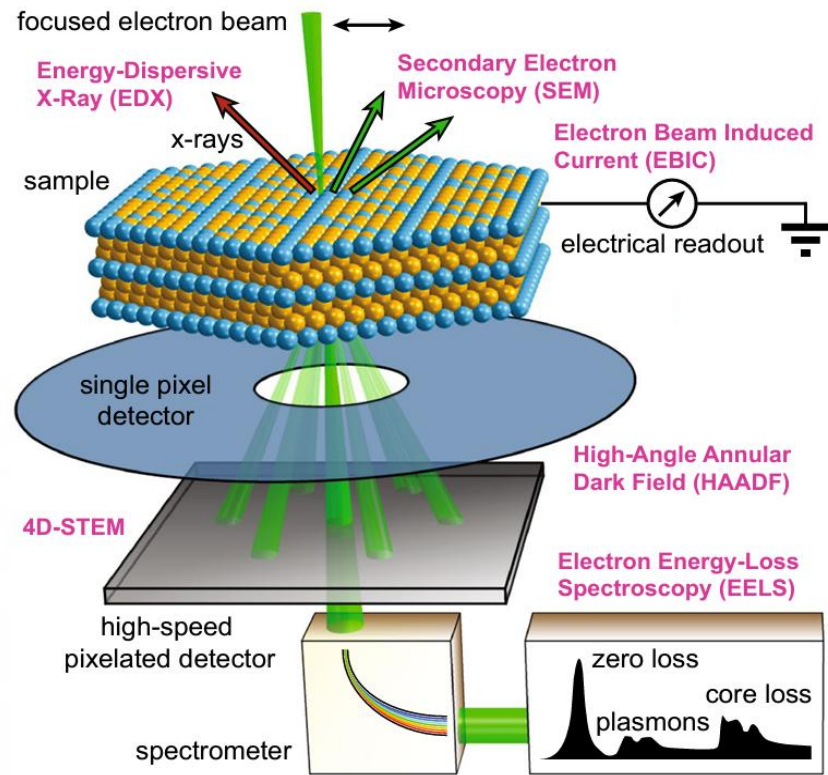
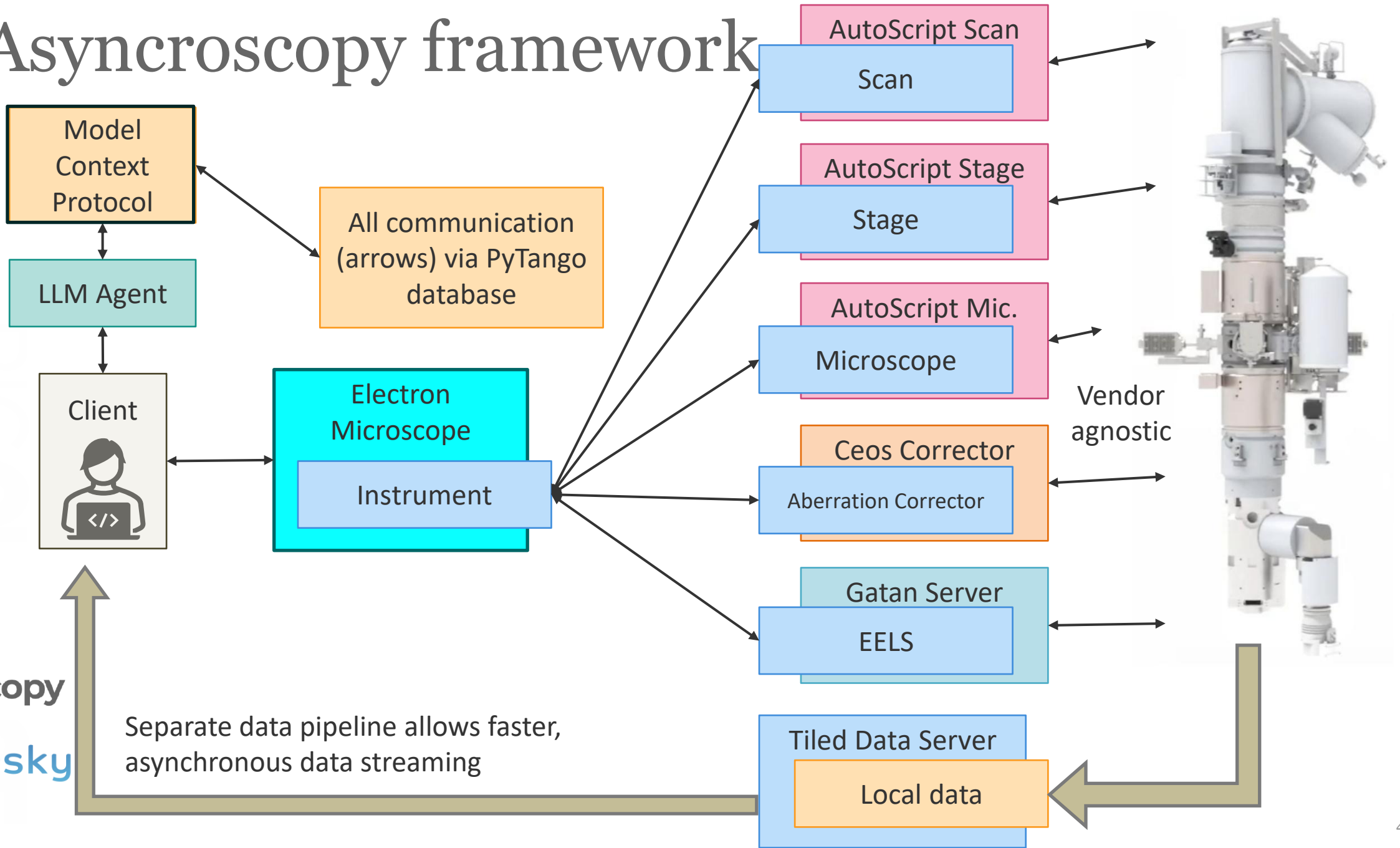


Image credit: Colin Opl... (LBNL), MRS Bulletin 2021

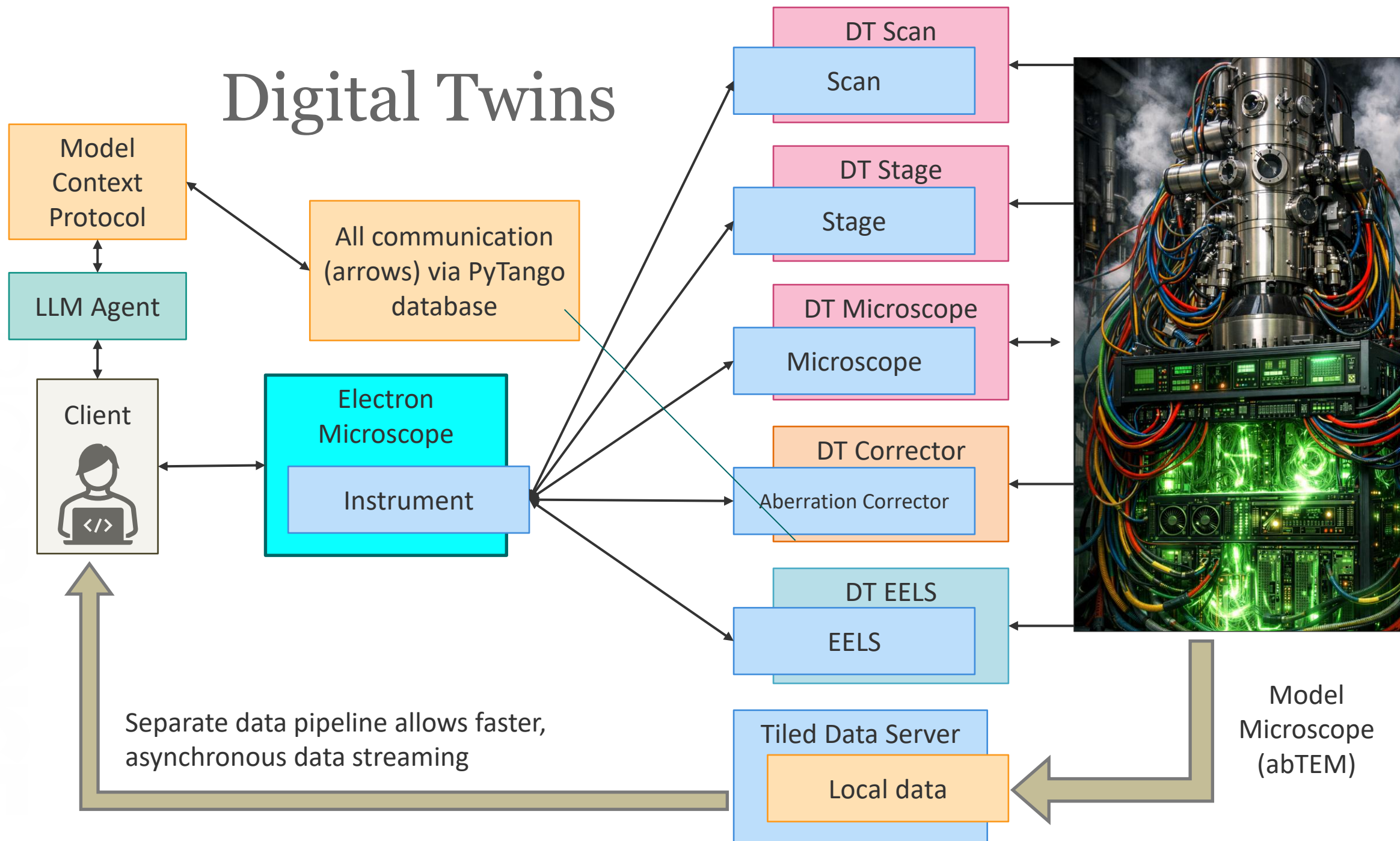
U. PRATIUSH ET AL, Rev. Sci. Instr. **95** (2024).

- Instrument has detectors and cameras from multiple manufacturers
- If you want to do anything useful, build APIs and orchestrators first!

Asyncroscopy framework



Digital Twins



Starting the servers

Asyncroscopy Server Startup

diffraction.yaml

Database

Tango host

Tango port

☒ Delete tango_database.db before start

Instrument

Instrument file

Hardware host

Hardware port

Timeout (seconds)

Device startup timeout

Data server

Devices

☒ camera ☐ corrector ☒ data ☒ eds ☒ scan ☒ stage

Terminal output

```

asyncroscopy.instruments.electron_microscope.detectors.eds eds_instance
scan 63189 0.0s uv run python -m
asyncroscopy.instruments.electron_microscope.hardware.scan scan_instance
stage 63190 0.0s uv run python -m
asyncroscopy.instruments.electron_microscope.hardware.stage stage_instance
instrument 63198 4.0s uv run python -m
asyncroscopy.instruments.electron_microscope.digital_twin_diffraction instrument_instance

TILED_URI http://127.0.0.1:9091
TILED_SERVING outputs/tiled_acquisitions

All asyncroscopy servers are ready.

Leave this terminal open while you use the servers. Press Ctrl+C to stop them.

```

Spectra300.yaml

```

configs > ! Spectra300.yaml
1 instrument:
2   class_name: AutoScriptMicroscope
3   file: asyncroscopy/instruments/electron_microscope/auto_script.py
4   description: "Thermo Fisher Spectra 300 TEM"
5   hardware_host: 10.46.217.241
6   hardware_port: 9095
7   timeout_seconds: 120
8 devices:
9   camera:
10    module_name: asyncroscopy.instruments.electron_microscope.detectors.camera
11   corrector:
12    module_name: asyncroscopy.instruments.electron_microscope.hardware.corrector
13   data:
14    module_name: asyncroscopy.data.data
15   eds:
16    module_name: asyncroscopy.instruments.electron_microscope.detectors.eds
17   scan:
18    module_name: asyncroscopy.instruments.electron_microscope.hardware.scan
19   stage:
20    class_name: AutoScriptSTAGE
21    module_name: asyncroscopy.instruments.electron_microscope.hardware.stage_autoscript
22    properties:
23      hardware_host: 10.46.217.241
24      hardware_port: 9095
25
26 tango:
27   host: 10.46.217.241
28   port: 9094
29   reset_database_file: false
30
31 tiled:
32   host: 10.46.217.241
33   port: 9091
34   acquisition_dir: outputs/tiled_acquisitions
35   autostart: true
36   register_on_startup: false
37
38 device_timeout_seconds: 120

```


Cloudified Instruments at UTK

- TEM/STEM Spectra 300, Thermo Fisher
- SPM Jupiter, Oxford Instruments
- SPM Cypher, Oxford Instruments
- Nano Indenter, KLA
- SPM/SEM FusionScope, QuantumDesign